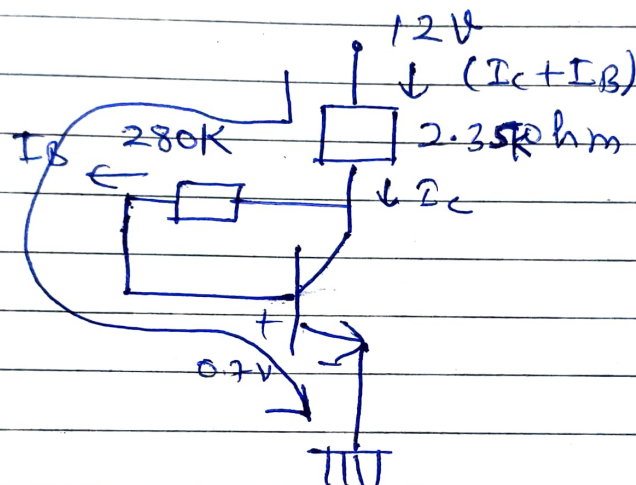
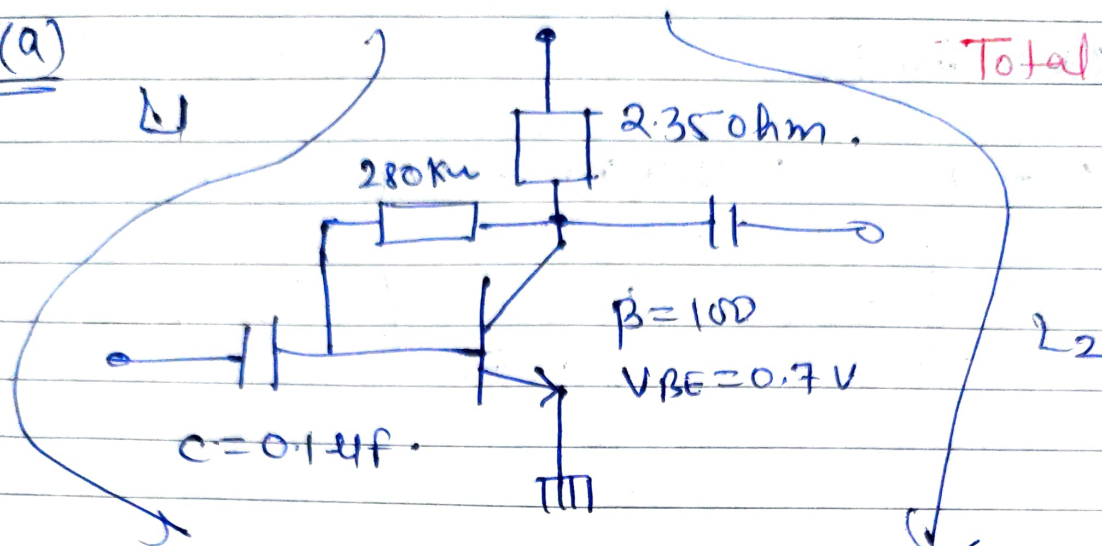


$$V_{CC} = 12V$$

Total $\rightarrow 2.5M$

Q2(a)



KVL in $L1$

$$12 - 2.35K \times (I_c + I_B) - 280K \times I_B - 0.7 = 0$$

$$I_B = \frac{12 - 0.7}{(\beta + 1) R_B + R_B} = \frac{12 - 0.7}{2.35K \times 101 + 280K}$$

$$I_B = \frac{11.3}{237.35K + 280K} = \frac{11.3}{517.35K} = 0.0218mA$$

I_B

Spiral

Date _____

$$I_B = 0.02184 \text{ mA} \text{ or } 21.84 \mu\text{A} \quad (0.5 \text{ M})$$

$$I_C = \beta I_B = 100 \times 21.84 \mu\text{A} = 2.18 \text{ mA}$$

1 M

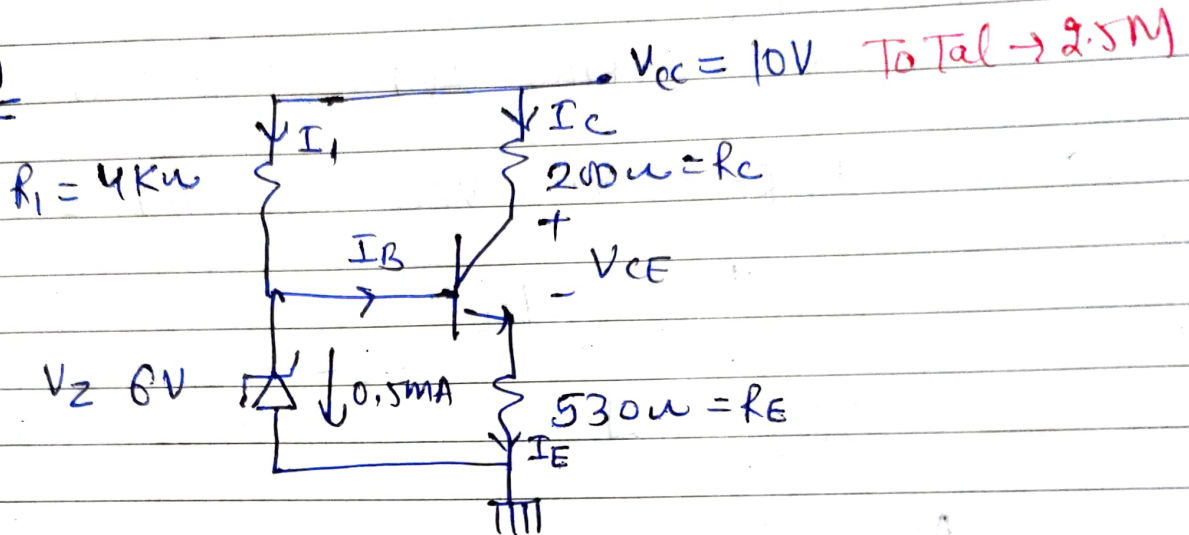
0

KVL in L_2

$$12 - 2.35 \text{ K} \times 2.2 \text{ mA} - V_{CE} = 0$$

$$V_{CE} = 6.83 \text{ V}$$

1 M

Mid sem Exam SolutionsQ.2(b)

for above ckt

$$I_B = \frac{V_{CC} - V_Z - I_Z}{R_1} = \frac{10 - 6 - 0.5}{4} = 0.5mA$$

$$\boxed{I_B = 0.5mA} \quad (0.5M)$$

Voltage across the emitter resistance,

$$V_E = V_Z - V_{BE} = 6 - 0.7 = 5.3V$$

Emitter current,

$$I_E = \frac{V_E}{R_E} = \frac{5.3}{530} - 0.5 = 10mA \quad (0.5M)$$

Collector current

$$\boxed{I_C = I_E - I_B = 10 - 0.5 = 9.5mA}$$

DC current gain of transistor,

$$\boxed{\beta = \frac{I_C}{I_B} = \frac{9.5}{0.5} = 19}$$

(1)

Spiral

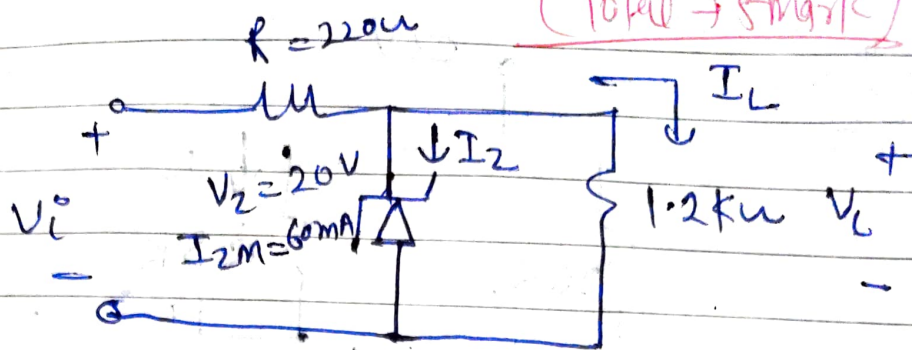
Date _____

~~Q. 10~~

$$\beta = \frac{\alpha}{1-\alpha} \quad \text{or}$$

$$\alpha = \frac{\beta}{1+\beta} = \frac{19}{20} = 0.95$$

(0.5 M)

Sol 3

$$V_{i\min} = \frac{(R_i + R) V_Z}{R_L} = \frac{(1200 + 220)(20)}{1200}$$

$$= 23.67 \text{ V} \quad (1.5 \text{ m})$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{20 \text{ V}}{1.2 \text{ K}} = 16.67 \text{ mA} \quad (1 \text{ m})$$

$$I_{R\max} = I_{ZM} + I_L = 60 \text{ mA} + 16.67 \text{ mA} = 76.67 \text{ mA} \quad (1 \text{ m})$$

$$\begin{aligned} V_{i\max} &= I_{R\max} R + V_Z \\ &= (76.67) \text{ mA} \times (0.22 \text{ K}) + 20 \text{ V} \\ &= 16.87 + 20 \\ &= 36.87 \text{ V} \quad (1.5 \text{ m}) \end{aligned}$$

Sol 4

Total → 5 mark.

solⁿ 4(a) DC: Testing $\beta R_E \geq 10 R_2$

$$(90)(1.5K\Omega) \geq 10(8.2K)$$

$$135K \geq 82K \text{ (satisfied)}$$

using approximation approach.

$$V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{8.2K(22V)}{56K + 8.2K} = 2.81V \quad (1M)$$

$$V_E = V_B - V_{BE} = 2.81V - 0.7V = 2.11V$$

$$I_E = \frac{V_E}{R_E} = \frac{2.11V}{1.5K} = 1.41mA \quad (1M)$$

$$r_e = \frac{26mV}{I_E} = \frac{26mV}{1.41mA} = 18.44\Omega \quad (1M)$$

$$b. R' = R_1 || R_2 = 56K || 8.2K = 7.15K\Omega$$

$$z_i = R' || \beta r_e = 7.15K || 90(18.44\Omega)$$

$$= 7.15K || 1.66K$$

$$= 1.35K$$

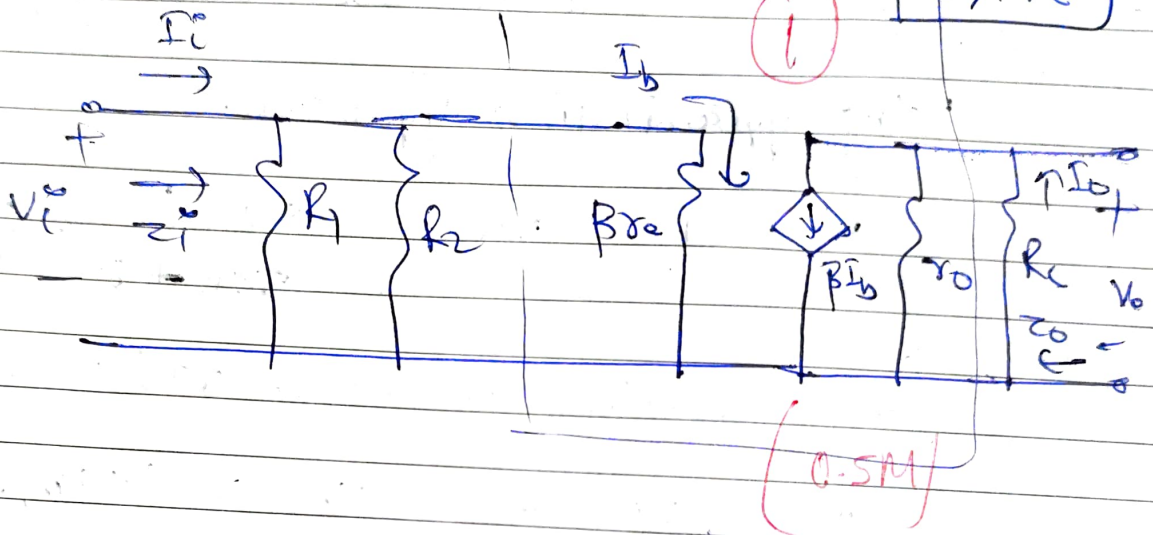
$$c. z_o = R_C = 6.8K \quad (1M)$$

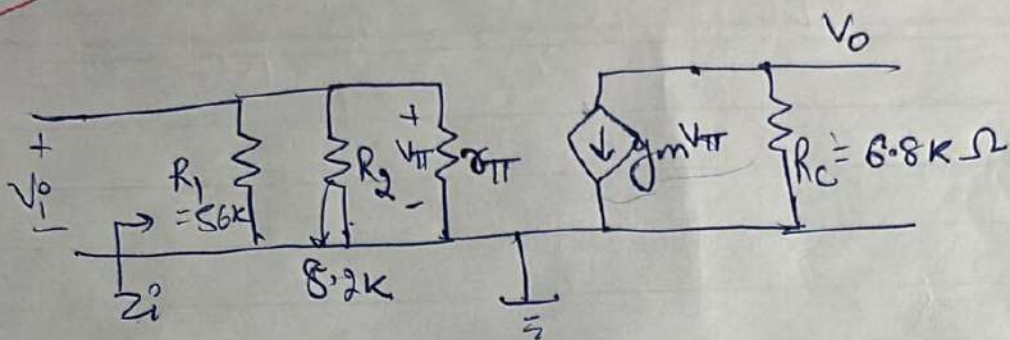
$$d. A_v = -\frac{R_C}{r_e} = \frac{6.8K}{18.44\Omega} = -368.76$$

$$e. \quad z_i = 1.35K\Omega$$

$$z_o = R_c \parallel r_o = 6.8K \parallel 50K = 5.98K \text{ vs } 6.8K\Omega$$

$$A_v = \frac{-R_c \parallel r_o}{r_e} = \frac{-5.98K}{18.44\mu} = -324.3 \text{ vs } -307.6$$





$$r_{\pi} = \frac{\beta}{g_m} = \frac{V_T}{I_B} = \frac{25\text{mV}}{0.015\text{mA}} = 1.66\text{k}\Omega$$

$$\approx \frac{26\text{mV}}{0.015\text{mA}} = 1.733\text{k}\Omega$$

Considered
 $I_B = 0.015\text{mA}$

$$r_{\pi} = \frac{\beta}{g_m} = \frac{I_C}{I_B}$$

$$g_m = \frac{I_C}{V_T} = \frac{1.394}{25} = 0.0557\text{V}^{-1}$$

$$\approx \frac{1.394}{26} = 0.0536\text{V}^{-1}$$

$-g_m R_C$

~~Ans~~ $V_o = -g_m V_{\pi} R_C$

$$V_i = V_{\pi}$$

$$\frac{V_o}{V_i} = -g_m R_C = -378.76 \left[V_T = 25\text{mV} \right]$$

$$= -364.58 \left[V_T = 26\text{mV} \right]$$